

P8106_HW2_SY3352

Su Yan

2026-03-15

```
library(caret)
library(tidymodels)
library(splines)
library(mgcv)
library(pdp)
library(earth)
library(tidyverse)
library(ggplot2)
```

Problem a

```
data <- read.csv("College.csv") |> na.omit()

set.seed(2026)
data_split <- initial_split(data, prop = 0.8)

training_data <- training(data_split)
testing_data <- testing(data_split)

fit_ss <- smooth.spline(training_data$Top10perc, training_data$Outstate)

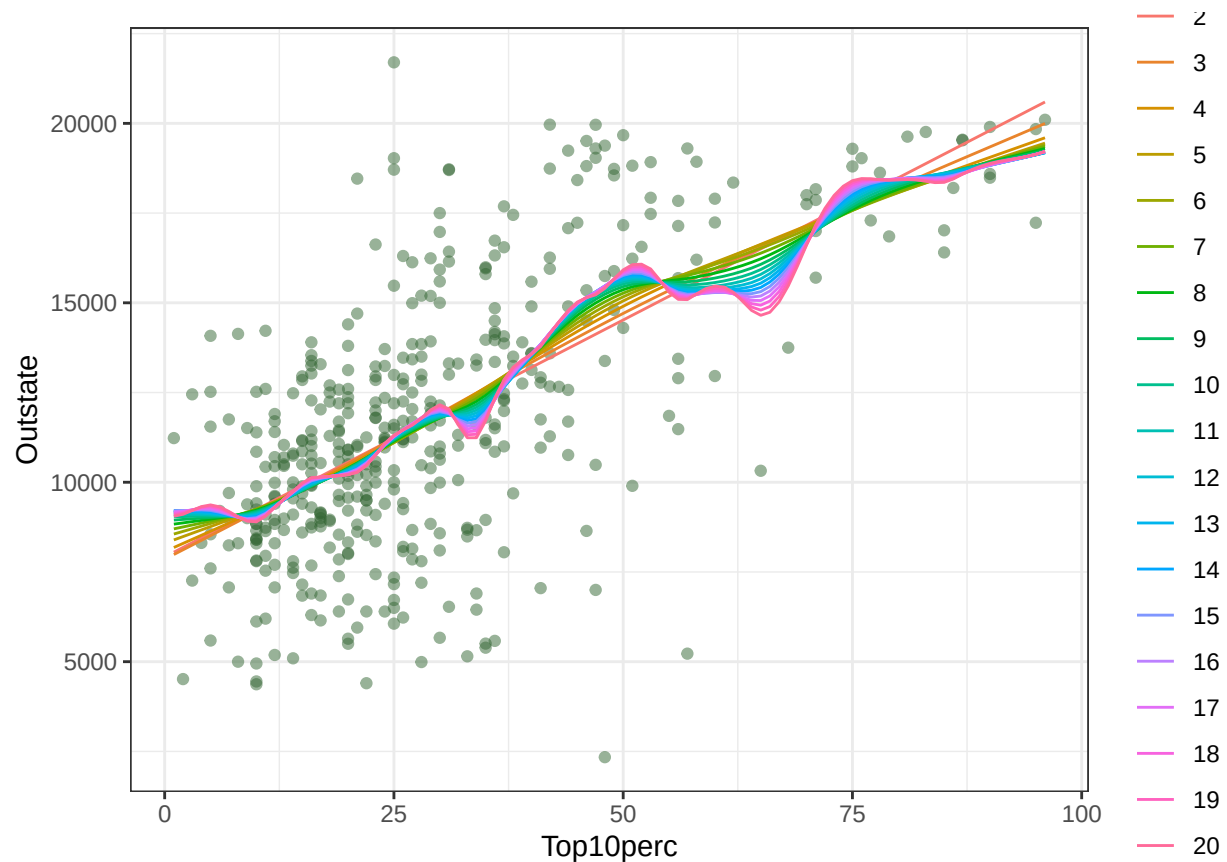
Top10perc.grid <- seq(from = min(training_data$Top10perc),
                      to = max(training_data$Top10perc), by = 1)
p <- ggplot(data = training_data, aes(x = Top10perc, y = Outstate)) +
  geom_point(color = rgb(.2, .4, .2, .5))

df_values <- seq(2, 20, by = 1)
pred_list = list()

for (df_ss in df_values) {
  fit <- smooth.spline(training_data$Top10perc, training_data$Outstate,
                      df = df_ss)
  pred <- predict(fit, x = Top10perc.grid)
  pred_list[[as.character(df_ss)]] <- data.frame(
    Top10perc1 = Top10perc.grid, Outstate1 = pred$y, df = as.factor(df_ss))
}

pred_df_all <- bind_rows(pred_list)

p + geom_line(aes(x = Top10perc1, y = Outstate1, color = df, group = df),
              data = pred_df_all) + theme_bw() + labs(color = "df")
```



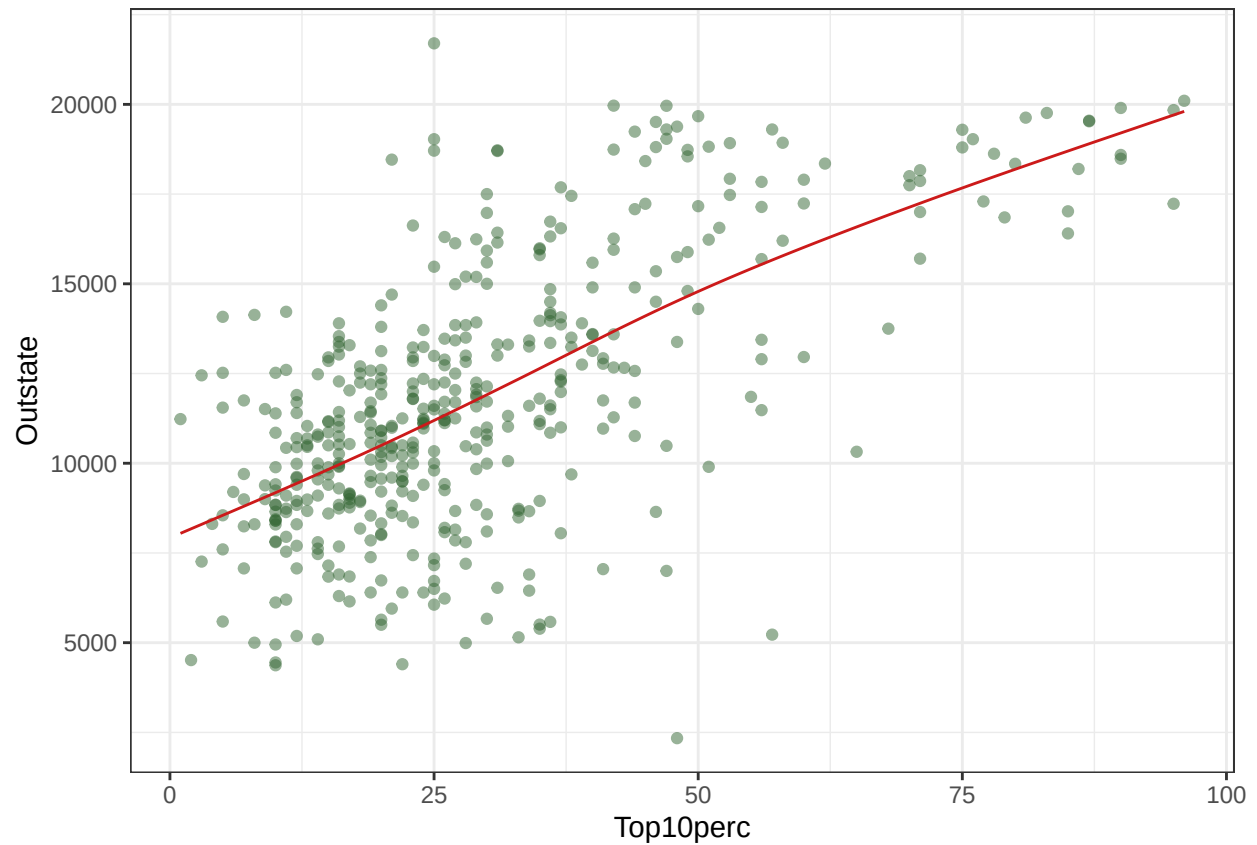
I fitted the degree of freedom from 2 to 20, it can be seen from the plot that as the degree of freedom go larger, the fitted curve become more wiggly and overfitting. With a low degree of freedom (2 to 4), the fitted curve is very smooth and almost straight, indicating that it might be underfitting and miss the non-linear pattern.

```
fit.ss$df
```

```
## [1] 3.399278
```

```
pred.ss <- predict(fit.ss, x = Top10perc.grid)
pred.ss.df <- data.frame(Top10perc1 = Top10perc.grid, Outstate1 = pred.ss$y)
```

```
p + geom_line(aes(x = Top10perc1, y = Outstate1),
               data = pred.ss.df,
               color = rgb(.8, .1, .1, 1)) + theme_bw()
```



The selected df is 3.3992784. Generalized cross-validation is used to select the degree of freedom because it efficiently approximates leave-one-out cross-validation error and is the standard approach for smoothing spline tuning.

Problem b

Problem c

Problem d

Problem e